CSE6214 SOFTWARE ENGINEERING FUNDAMENTALS
MARCH/APRIL 2026 (TERM2610)
PROJECT DESCRIPTION

Significant Dates:

- Group formation : 30.03.2026 – 12.04.2026 (Week 01 & 02)
- Submission Project
  - Part I (Project Planning / Requirements Analysis) : 01.05.2026, 12 midnight (Week 05)
  - Part II (Design / Architecture / Interfaces / Database) : 05.06.2026, 12 midnight (Week 10)
  - Part III (Development / Testing / Project Monitoring & Reporting) : 26.06.2026, 12 midnight (Week 13)
- Presentation : 29.06.2026 – 03.07.2026 (Week 14)

* Please strictly adhere to the important dates above.
* * A kind reminder for every students that **penalty for late submission** could be applied.

Instructions:

Students need to form a group of **3** to 4 from the **SAME TUTORIAL** section. Students are required to produce and submit documentation on requirements, design and implementation (prototype) of a system. Project rubrics for Part I, Part II and Part III are detailed on pages 8, 9, and 10, respectively. The project titles are assigned based on tutorial sections, listed as follows:

Tutorial Section	Project Title	Key Processes

TT1L
Dr. Naveen
	Campus Resource Booking System	Users: 3-4 Users (eg: Student / Faculty or Staff / Resource Manager / admin)
		• User Management
		  ○ Account registration and login functionality
		  ○ Store user details, department affiliation, and contact information
		  ○ Profile management with resource assignments and booking history
		• Resource Management
		  ○ Register resources (meeting rooms, labs, projectors, AV equipment)
		  ○ Resource details: capacity, features, location, availability calendar
		  ○ Equipment specifications (HDMI, wireless mic, whiteboard, etc.)
		  ○ Maintenance schedules and status (available, maintenance, faulty)
		• Booking Management

		○ Search resources by type, location, date/time, capacity ○ Create bookings with start/end times, purpose, number of attendees ○ Recurring booking support (weekly classes, regular meetings) ○ Booking approval workflow for specialized equipment ○ Modify/cancel bookings within policy timeframes (24 hours notice) ○ Auto-cancel bookings with no-show detection • Reporting & Notifications ○ Dashboard with calendar/matrix view of bookings ○ Utilization reports (peak hours, no-show rates) ○ Resource utilization heatmaps by time/day/department ○ Booking trend analysis and compliance reports ○ Email confirmations, reminders, and conflict alerts ○ Maintenance notifications affecting scheduled bookings
TT2L **Dr. Naveen**	Campus Job Board System	Users: 3-4 Users (eg: Students / Job Posters / Hiring Supervisors / Admins • User Management ○ Account registration, login, and role-based access control ○ Email verification for account activation ○ Store user details, department affiliation, contact information, and skills ○ Availability calendar, resume upload, job preferences, and posting limits • Job Management ○ Create job postings with title, department, pay rate, required skills ○ Specify job duration, shift patterns, application deadlines ○ Upload job description documents and auto-generate job IDs ○ Application submission with resume, cover letter, and availability • Workflow & Tracking ○ Application status tracking (Draft, Submitted, Shortlisted, Interview Scheduled, Hired, Rejected) ○ Skills-based application ranking and shortlisting ○ Schedule interviews with calendar integration ○ Update candidate status, provide feedback, approve timesheets • Reporting & Notifications ○ Job posting response rates, fill rates, hiring trends, time-to-hire metrics ○ Employment statistics by skillset and department ○ Email notifications for job postings, application updates, interviews ○ Deadline reminders, low response warnings, budget threshold alerts
TT3L **Dr. Mohana**	Smart Study Planner & Task Management System	Users (3–4 users) (e.g. Student, Lecturer, Academic Advisor, Admin) • User Management ○ Account registration, login, and role-based access control ○ Email verification for account activation ○ Store user details, enrolled courses, academic goals, and study preferences ○ Personal dashboard setup (study hours, preferred schedule, productivity style)

		• Study & Task Management ○ Create and manage study plans (subjects, topics, exam dates) ○ Add tasks such as assignments, quizzes, revision sessions ○ Upload supporting materials (notes, links, documents) ○ Auto-generate task IDs and organize by course/module ○ Set priorities, estimated study duration, and deadlines • Workflow & Tracking ○ Task status tracking (Planned, In Progress, Completed, Overdue) ○ Automated study schedule generation based on deadlines and priorities ○ Progress tracking (completion %, time spent vs planned) ○ Lecturer/Advisor input (recommended study plans or adjustments) ○ Calendar integration for study sessions and academic events • Reporting & Notifications ○ Study performance insights (task completion rate, consistency trends) ○ Time management analytics (planned vs actual study hours) ○ Course-based progress reports ○ Email/app notifications for deadlines, overdue tasks, and reminders ○ Alerts for poor progress or heavy workload periods
TT4L **Dr. Mohana**	Digital Fitness Coaching & Activity Tracker	Users: 3–4 Users (e.g.: User / Fitness Coach / Health Advisor / Admin) • User Management ○ Account registration, login, and role-based access control ○ Email verification for account activation ○ Store user details (age, weight, fitness level, health goals) ○ Profile setup (preferred workouts, dietary preferences, availability schedule) • Fitness & Activity Management ○ Create personalized workout plans (by coach or system-generated) ○ Log daily activities (workouts, steps, calories burned) ○ Upload fitness data (images, progress photos, wearable device sync) ○ Set fitness goals (weight loss, muscle gain, endurance) ○ Auto-generate activity/session IDs and history logs • Workflow & Tracking ○ Activity status tracking (Planned, Ongoing, Completed, Missed) ○ Progress monitoring (weight, BMI, strength, endurance metrics) ○ Coach feedback and plan adjustments ○ Goal tracking with milestone achievements ○ Schedule workouts with calendar integration • Reporting & Notifications ○ Performance analytics (weekly/monthly fitness progress) ○ Goal achievement reports and activity consistency trends ○ Health insights (calories burned, workout frequency) ○ Notifications for workout reminders, inactivity alerts, and milestones ○ Alerts for missed sessions or declining performance

<table>
<tr><td>TT5L
Dr. Chua Fang Fang</td><td>Campus Food Ordering and Management System</td><td>Users: 3–4 Users (e.g.: Student/Vendor/Admin)
User Registration & Login,
Vendor Management,
Food Ordering Management (i.e. add items to cart),
Order Scheduling (i.e. pre-order, pick-up time selection),
Order Tracking,
Payment,
Queue Management,
Ratings & Review,
Analytics & Reporting</td></tr>
<tr><td>TT6L
Mr.Shaari</td><td>Digital Mental Wellness Support System</td><td>Users: 3–4 Types (Student / Counselor / Wellness Officer / Admin)
User Management :
Account Registration & Login, Account registration and login functionality, Role-based access (student, counselor, wellness officer, admin).
Profile Management: Store biodata, faculty/department, demographic details, Mental wellness profile (self-assessment history, session history) , Students can update personal details and manage privacy settings.
Counselor Profiles: Expertise areas (stress management, anxiety, academic counseling), Work schedule and availability tracking
Digital Wellness Assessment Module
Standardized Self-Assessment Tests: Stress Level Check, Mood Tracking (daily/weekly logs), Anxiety Self-Rating Scales
Assessment Engine: Auto-generate scores with interpretation categories (low/medium/high)
Historical & Trend Tracking: View mood and stress trends over time (graphs stored from database), Weekly/monthly wellness summary reports
Session & Appointment Management
Search & Schedule Counseling Sessions: Filter by counselor, availability, counseling type (academic, emotional, general wellness)
Booking System: Real-time slot availability, Student can book, modify, or cancel appointments, Auto-prevention of double-booking
Counselor Dashboard: View upcoming sessions, Update session notes or recommendations, Track follow-up sessions
Wellness Content Management
Digital Resource Library (Articles, videos, coping strategies, relaxation techniques)
Admin/Wellness Officer Controls (Upload or categorize content by topic - stress, sleep hygiene, mental resilience)
Reporting & Notifications
Student Dashboard (Stress/mood trend charts, Upcoming session reminders)
Counselor Dashboard (Session calendar view, Student assessment summaries)</td></tr>
</table>

		○ Automated Notifications (Email/SMS reminders for session booking)
TT7L **Mr. Shaari**	Smart Interactive Learning System	Users: 3–4 Types (Student / Instructor / Academic Advisor / Admin) • User Management ○ Account Registration & Login, Account registration and login functionality, Role-based access (student, instructor, academic advisor, admin) ○ Profile Management: Store academic level, programme/department, learning preferences, Instructor profiles: specialization, subjects taught, office hours ○ User Activity Tracking: Log learning activities: quizzes taken, videos watched, pages visited • Course & Content Management ○ Course Creation & Management: Instructors create courses with modules, lessons, assessments ○ Interactive Learning Materials: Embedded videos, Interactive quizzes (Automated feedback for each quiz attempt, tips for improvement based on mistakes) • Assessment & Progress Tracking ○ Quiz & Assignment Management: Multiple question types: MCQ, fill-in-the-blank, short answer, Randomized question bank generation, Auto-grading ○ Progress Tracking: Show module completion status, Detailed performance analytics ○ Student and Instructor Dashboard: Personalized learning progress, Recommended next steps, Past grades and assessment history • Reporting & Notifications ○ Analytics Dashboard (Course engagement charts, summary of Student performance distribution) ○ Notifications (Assignment deadlines, new course content uploaded, quiz score announcements)

Note: Name your file in the following format:

Proj_Px_TTxL_Gx_stud1, stud2, stud3, stud4.pdf

Example:

Proj_Px_TT3L_G5_David, Edwin, Ella, Othman.pdf
Proj: obviously this is Project.
Px: Project part: PI,PII, or PIII.
TTxL: your tutorial class code: TT1L, TT2L, TT3L, etc.
Gx: your group number.
stud: group member's name (short one, EX:Studname: **David** Fong Seow Kee)

CSE6214 SOFTWARE ENGINEERING FUNDAMENTALS
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PROJECT RUBRIC

Lecturer Name : ______________________ **Tutorial Section** : ______________________

Project Title : ______________________ **Presentation Date** : ______________________

Project Phase	Cognitive	Affective	Total
Project - Part 1 (20%)		NIL	
Project - Part II (20%)		NIL	
Project - Part III (30%)	NIL		
Grand Total (70%)			

Signed by:

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(Lecturer's Name and Date)

CSE6214 SOFTWARE ENGINEERING FUNDAMENTALS
MARCH/APRIL 2026 (TERM2610)
COGNITIVE AND AFFECTIVE ASSESSMENT

Mark Distribution	Submission Type	Descriptive Elements	Weightage	Rate (0-5)	Total (Weightage * Rate)
Cognitive and Affective Components	**Project – Part I Project Planning / Requirements Analysis (Documentation)**	1) System Overview describes users, scenario, problems, & use cases. Use case diagram and descriptions reflects System Overview details. The process flow is clear in each use case.	1		
		2) Class diagram/ER diagram have the main entities.	1		
		3) All processes are related and integrated. Additional functions and innovations for a more complete and usable software.	1		
		4) Quality and correctness of Gantt chart, diagrams and notations / Clearly and coherently written academic discourse.	1		
	Project –Part II Design / Architecture / Interfaces / Database (Documentation)	1) Database design matches requirements.	1		
		2) Architecture design shows the structure of the solution. Additional modules and innovations for a more complete and usable software.	1		
		3) Processing described in sequence diagrams, activity diagrams, state transition diagrams and interface design matches the requirements.	1		
		4) Quality and correctness of diagrams and notations / Clearly and coherently written academic discourse.	1		
	Project - Part III Development / Testing / Project Monitoring & Reporting (Documentation + Video + Presentation)	1) Quality of Prototype System/App etc.(Software Development Outcome)	1		
		2) Software Testing Procedures and Strategies (Testing)	1		
		3) Quality and correctness of diagrams and notations / Clearly and coherently written academic discourse.	1		
		4) Work Responsibility and Work Relations towards updated Requirements & Design - (Group Relations + Documentation)	1		
		5) Video - Screens & Explanation, Linking & Flow of system	1		
		6) Presentation - Clear Delivery of Ideas	1		
		Project I = Total (Weightage * Rate)		**Max (20%)**	
		Project II = Total (Weightage * Rate)		**Max (20%)**	
		Project III = Total (Weightage * Rate)		**Max (30%)**	
		GRAND TOTAL =		**Max (70%)**	

Note for Rate: 0-non existence, 1-very weak, 2-weak, 3-fair, 4-good, 5-excellent

COGNITIVE COMPONENT RUBRIC
PROJECT - PART I – PROJECT PLANNING / REQUIREMENTS ANALYSIS

Descriptive Elements	**Very Weak (1)**	**Weak(2)**	**Fair(3)**	**Good(4)**	**Excellent(5)**
1) System Overview describes users, scenario, problems, & use cases. Use case diagram and descriptions reflects System Overview details. The process flow is clear in each use case.	System Overview is very ambigious, unresonable users, unreasonable scenarios, and insufficient user cases. Use Case diagram and descriptions that solves only *5% of the problems*	System Overview is ambigious, number of users is doubful, unreasonable scenarios, and insufficient user cases. Use Case diagram and descriptions that solves *25% of the problems*	System Overview is almost clear, number of users, fair description of scenarios, and user cases. Use Case diagram and descriptions that solves *50% of the problems*	System Overview is clear, number of users, some good scenarios, and good user cases. Good Use Case diagram and descriptions that solves *80% of the problems*	System Overview is very clear, number of users, concrete scenarios, and concrete user cases. Comprehensive and complete Use Case diagram and descriptions that solves all problems. Very good self-explanatory diagrams.
2) Class diagram/ER diagram have the main entities	Class / ER diagrams have only *5% of the main entities*, diagrams are not understandable at all.	Class / ER diagrams have *25% of the main entities*, diagrams are not easy to understand.	Class / ER diagrams have *50% of the main entities*, self-explanatory diagrams.	Class / ER diagrams have *80% of the main entities*, good self-explanatory diagrams.	Class / ER diagrams have all the main entities, very good self-explanatory diagrams.
3) All processes are related and integrated. Additional functions and innovations for a more complete and usable software.	Processes are disparate and not integrated, basic functions with no innovations.	Processes are somewhat related but not well integrated, with little or no innovations.	Processes are integrated with some minor additional functions/innovation.	Processes are integrated with some additional useful functions/innovations.	Well integrated processes, with good additional functions and innovations for a complete and usable software.
4) Quality and correctness of Gantt chart, diagrams and notations / Clearly and coherently written academic discourse	Not able to draw Gantt chart, diagrams and write ideas clearly and coherently	Able to draw Gantt chart, diagrams and write ideas with limited clarify and coherence and require further improvements	Able to draw Gantt chart, diagrams and write ideas fairly coherently and clearly but require minor improvements	Able to draw Gantt chart, diagrams and write ideas coherently and clearly	Able to draw Gantt chart, diagrams and write ideas with excellent coherence and clarity

Note for Rate: 0 = non existence

COGNITIVE COMPONENT RUBRIC
PROJECT - PART II – DESIGN / ARCHITECTURE / INTERFACES / DATABASE

Descriptive Elements	Very Weak (1)	Weak(2)	Fair(3)	Good(4)	Excellent(5)
1) Database design matches requirements	Database design *matches 5%* of the requirements, diagrams/designs are not understandable at all.	Database design *matches 25 %* of the requirements, diagrams/designs are not easy to understand.	Database design *matches 50%* of the requirements, self-explanatory diagrams/designs.	Database design *matches 80%* of the requirements, good self-explanatory diagrams/designs.	Database design *matches all the requirements,* very good self-explanatory diagrams/designs.
2) Architecture design shows the structure of the solution. Additional modules and innovations for a more complete and usable software.	Architecture design shows *5% structure of the solutions,* basic modules with no innovations and diagrams/designs are not understandable at all.	Architecture design shows *25% structure of the solutions,* with little or no innovations.and diagrams/designs are not easy to understand.	Architecture design shows *50% structure of the solutions,* with some minor additional modules/innovation and self-explanatory diagrams/designs.	Architecture design shows *80% structure of the solutions,* with some additional useful modules/innovations and good self-explanatory diagrams/designs.	Architecture design shows *all the structure of the solutions,* with good additional modules and innovations, and very good self-explanatory diagrams/designs.
3) Processing described in sequence diagrams, activity diagrams, state transition diagrams and interface design matches the requirements.	Sequence diagrams, activity diagrams, state transition diagrams and interface design matches *5% of the requirements,* diagrams/designs are not understandable at all.	Sequence diagrams, activity diagrams, state transition diagrams and interface design matches *25% of the requirements,* diagrams/designs are not easy to understand.	Sequence diagrams, activity diagrams, state transition diagrams and interface design matches *50% of the requirements,* self-explanatory diagrams/designs.	Sequence diagrams, activity diagrams, state transition diagrams and interface design matches *80% of the requirements*, good self-explanatory diagrams/designs.	Sequence diagrams, activity diagrams, state transition diagrams and interface design matches *all the requirements,* very good self-explanatory diagrams/designs.
4) Quality and correctness of diagrams and notations / Clearly and coherently written academic discourse	Not able to draw diagrams and write ideas clearly and coherently	Able to draw diagrams and write ideas with limited clarify and coherence and require further improvements	Able to draw diagrams and write ideas fairly coherently and clearly but require minor improvements	Able to draw diagrams and write ideas coherently and clearly	Able to draw diagrams and write ideas with excellent coherence and clarity

Note for Rate: 0 = non existence

AFFECTIVE COMPONENT RUBRIC
PROJECT PART III – DEVELOPMENT / TESTING / PROJECT MONITORING & REPORTING

Descriptive Elements	Very Weak (1)	Weak(2)	Fair(3)	Good(4)	Excellent(5)
1) Quality of Prototype System/App etc.(Software Development Outcome)	Quality of Prototype System/App is very weak, and able to *demonstrate only 5%* of the required functionalities.	Quality of Prototype System/App is weak, and able to *demonstrate 25%* of the required functionalities.	Quality of Prototype System/App is fair, and able to *demonstrate 50%* of the required functionalities.	Quality of Prototype System/App is good, and able to *demonstrate 80%* of the required functionalities.	Quality of Prototype System/App is excellent, and able to *demonstrate all* the required functionalities.
2) Software Testing Procedures and Strategies (Testing)	Software Testing Procedures and Strategies is *demonstrated,* but the intent of finding *particular errors is very shallow/weak.*	Software Testing Procedures and Strategies is *demonstrated,* but the intent of *finding particular errors is shallow/weak.*	Software Testing Procedures and Strategies is *demonstrated, but in general, with unspecific intent* of finding errors.	Software Testing Procedures and Strategies is *demonstrated, but in general, with a specific* intent of finding particular errors.	Software Testing Procedures and Strategies is *demonstrated in detail, with a specific intent* of finding particular errors.
3) Quality and correctness of diagrams and notations. / Clearly and coherently written academic discourse.	Not able to draw diagrams and write ideas clearly and coherently	Able to draw diagrams and write ideas with limited clarify and coherence and require further improvements	Able to draw diagrams and write ideas fairly coherently and clearly but require minor improvements	Able to draw diagrams and write ideas coherently and clearly	Able to draw diagrams and write ideas with excellent coherence and clarity
4) Work Responsibility and Work Relations towards updated Requirements & Design - (Group Relations + Documentation)	Does not perform assigned tasks within the scope of work even with close supervision / Has a disharmonious relationship between group members resulting in poorly coordinated work	Perform assigned tasks within by the scope of work with close supervision / Has a less harmonious relationship between group members resulting in weakly coordinated work	Perform assigned tasks within by the scope of work and meets expectation / Has a satisfactory relationship between group members resulting in coordinated work at acceptable level	Perform assigned tasks within by the scope of work and exceeds expectation / Has a good relationship between group members resulting in good coordinated work	Perform assigned tasks beyond the scope of work and beyond expectation / Has a well-acknowledged relationship between group members resulting in excellent coordinated work
5) Video – Screens & Explanation, Linking & Flow of system	Video content has *5% of the screens, explanations, linking and flow* of the system, visual video contents are not understandable at all.	Video content has *25% of the screens, explanations, linking and flow* of the system, visual video contents are not easy to understand.	Video content has *50% of the screens, explanations, linking and flow* of the system, self explanatory visual video contents.	Video content has *80% of the screens, explanations, linking and flow* of the system, good clear and self explanatory visual video contents.	Video content has *all the screens, explanations, linking and flow* of the system, very clear and self explanatory visual video contents.
6) Presentation - Clear Delivery of Ideas	Not able to deliver ideas clearly and require major improvements	Able to deliver ideas and require further improvements	Able to deliver ideas fairly clearly and require minor improvements	Able to deliver ideas clearly	Able to deliver ideas with great clarity

Note for Rate: 0 = non existence

USEFUL LINKS FOR PROJECT

Use Case links

https://www.uml-diagrams.org/use-case-diagrams.html
https://www.uml-diagrams.org/use-case-diagrams-examples.html
https://www.lucidchart.com/pages/uml-use-case-diagram
https://www.smartdraw.com/use-case-diagram/
https://online.visual-paradigm.com/tutorials/use-case-diagram-tutorial/

video

https://www.youtube.com/watch?v=zid-MVo7M-E

Activity diagram links

https://www.uml-diagrams.org/activity-diagrams-examples.html
https://www.lucidchart.com/pages/uml-activity-diagram
https://www.lucidchart.com/pages/swimlane-diagram
https://www.smartdraw.com/activity-diagram/examples/
https://www.visual-paradigm.com/guide/uml-unified-modeling-language/what-is-activity-diagram/

video

https://www.youtube.com/watch?v=yAihwmczqsk

ER diagram links

https://www.smartdraw.com/entity-relationship-diagram/
https://www.smartdraw.com/entity-relationship-diagram/examples/
https://creately.com/blog/diagrams/er-diagrams-tutorial/
http://www.cs.uregina.ca/Links/class-info/215/erd/

video

https://www.youtube.com/watch?v=QpdhBUYk7Kk
https://www.youtube.com/watch?v=-CuY5ADwn24
https://www.youtube.com/watch?v=c0_9Y8QAstg
https://www.youtube.com/watch?v=-fQ-bRllhXc

Class diagrams links

https://www.lucidchart.com/pages/uml-class-diagram
https://www.tutorialspoint.com/uml/uml_component_diagram.htm
https://www.smartdraw.com/class-diagram/
https://www.ibm.com/developerworks/rational/library/content/RationalEdge/sep04/bell/index.html
https://www.visual-paradigm.com/guide/uml-unified-modeling-language/what-is-class-diagram/

videos

https://www.youtube.com/watch?v=UI6lqHOVHic
https://www.youtube.com/watch?v=xiUFTLIU-Iw
https://www.youtube.com/watch?v=ZyST6OFtb7k

State transition diagram links

http://www.cs.unc.edu/~stotts/145/CRC/state.html
https://www.stickyminds.com/article/state-transition-diagrams

https://www.smartdraw.com/state-diagram/
https://www.lucidchart.com/pages/uml-state-machine-diagram

videos

https://www.youtube.com/watch?v=PF9QcYWIsVE
https://www.youtube.com/watch?v=OsmWASXE2IM

Sequence diagram links

https://www.lucidchart.com/pages/uml-sequence-diagram
https://www.smartdraw.com/sequence-diagram/
https://www.visual-paradigm.com/guide/uml-unified-modeling-language/what-is-sequence-diagram/
https://www.ibm.com/developerworks/rational/library/3101.html

videos

https://www.youtube.com/watch?v=XIQKt5Bs7II
https://www.youtube.com/watch?v=cxG-qWthxt4
https://www.youtube.com/watch?v=18_kVIQMavE